

8.16

a. Use the data file vacation to estimate the model by OLS. Construct a 95% interval estimate for the effect of one more child on miles traveled, holding the two other variables constant.

● R code :

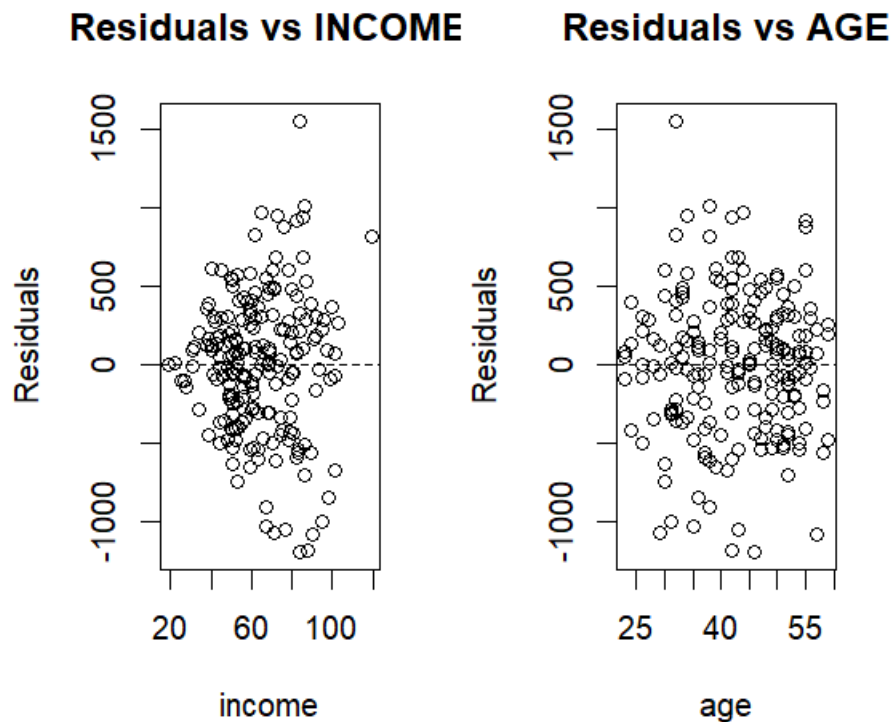
```
> #(a)
> alpha = 0.05
> mod1 = lm(miles ~ income + age + kids ,data = vacation)
> smod1 = summary(mod1)
> df = df.residual(mod1)
> b4 = coef(mod1)[[4]]
> tc = qt(1-alpha/2, df)
> se_b4 = coef(smod1)[4,2]
> uc = b4 + tc * se_b4
> lc = b4 - tc * se_b4
> cat(
+   "8.16(a)", "\n",
+   "the 95% CI is : [",lc,"",uc,"]", "\n",
+   sep=""
+ )
8.16(a)
the 95% CI is : [-135.3298,-28.32302]
```

● Answer: The 95% CI for the effect of one more child on miles traveled is [-135.3298 , -28.32302]

b. Plot the OLS residuals versus INCOME and AGE. Do you observe any patterns suggesting that heteroskedasticity is present?

● R code :

```
> #(b)
> par(mfrow = c(1, 2)) #「將繪圖視窗分割成 1 列 2 欄，讓接下來畫的兩張圖可以左右併排顯示。」
> plot(vacation$income, resid(mod1),
+      main = "Residuals vs INCOME",
+      xlab = "income", ylab = "Residuals")
> abline(h = 0, lty = 2) #在現有的圖表上增加一條水平線，通常用作「基準線」，觀察數據點是否有偏離。
> plot(vacation$age, resid(mod1),
+      main = "Residuals vs AGE",
+      xlab = "age", ylab = "Residuals")
> abline(h = 0, lty = 2)
```



- Answer : 可以看到左圖的 residual 間的差距隨 Income 增加而增加，因此有 heteroskedasticity；右圖的 residual 間的差距沒有特定的趨勢在，因此沒有 heteroskedasticity

c. Sort the data according to increasing magnitude of income. Estimate the model using the first 90 observations and again using the last 90 observations. Carry out the Goldfeld-Quandt test for heteroskedastic errors at the 5% level. State the null and alternative hypotheses.

- R code :

```

> vacation_sorted = vacation[order(vacation$income), ]
> low90 = vacation_sorted[1:90, ]
> high90 = vacation_sorted[111:200, ]
>
> mod_low <- lm(miles ~ income + age + kids, data = low90)
> mod_high <- lm(miles ~ income + age + kids, data = high90)
>
> sse_low <- sum(resid(mod_low)^2)
> sse_high <- sum(resid(mod_high)^2)
> df_low = df.residual(mod_low)
> df_high = df.residual(mod_high)
> F_stat <- (sse_high / df_high) / (sse_low / df_low)
> fc_u = qf(1-alpha/2, df_high, df_low)
> fc_l = qf(alpha/2, df_high, df_low)
> cat(
+   "8.16(c)", "\n",
+   "the F* is : ", F_stat, "\n",
+   "and fc_u is : ", fc_u, "\n",
+   "and fc_l is : ", fc_l, "\n",
+   sep=""
+ )
8.16(c)
the F* is : 3.104061
and fc_u is : 1.530373
and fc_l is : 0.6534355

```

● Answer :

$H_0 : \sigma^2(\text{low}) = \sigma^2(\text{high})$

$H_a : \sigma^2(\text{low}) \neq \sigma^2(\text{high})$

GQ Test' s F-stat(F^*) = 3.104061

Critical value_upper = 1.530373

Critical value_lower = 0.6534355

因為 $F^* = 3.104061 > \text{critical value_upper} = 1.530373$ ，所以我們可以拒絕 $H_0 : \sigma^2(\text{low}) = \sigma^2(\text{high})$ ，推論 $H_a : \sigma^2(\text{low}) \neq \sigma^2(\text{high})$ 為真

d. Estimate the model by OLS using heteroskedasticity robust standard errors. Construct a 95% interval estimate for the effect of one more child on miles traveled, holding the two other variables constant. How does this interval estimate compare to the one in (a)?

● R code :

```

> mod1_hc = hccm(mod1,type = "hc1")
> vac.HC1 = coeftest(mod1, vcov.=mod1_hc)
> b4 = vac.HC1[4,1]
> se_b4_rob = vac.HC1[4,2]
> uc_rob = b4 + tc * se_b4_rob
> lc_rob = b4 - tc * se_b4_rob
>
> cat(
+   "8.16(d)", "\n",
+   "the 95% CI using heteroskedasticity robust standard errors is : [",lc_rob,";",uc_rob,"]", "\n",
+   sep=""
+ )
8.16(d)
the 95% CI using heteroskedasticity robust standard errors is : [-139.323,-24.32986]

```

● Answer :

The 95% confidence interval using heteroskedasticity robust standard errors is [-139.323, -24.32986].

與(a)小題的 95% confidence interval 相比，信賴區間變得比較寬

- e. Obtain GLS estimates assuming $\sigma_i^2 = \sigma^2 INCOME_i^2$. Using both conventional GLS and robust GLS standard errors, construct a 95% interval estimate for the effect of one more child on miles traveled, holding the two other variables constant. How do these interval estimates compare to the ones in (a) and (d)?

● R code :

```

> #(e)
> # Conventional GLS
> w = 1/vacation$income^2
> mod1.wls = lm(miles ~ income + age + kids, weights=w, data=vacation)
> smod1.wls = summary(mod1.wls)
> se_b4.wls = coef(smod1.wls)[4,2]
> b4.wls = coef(mod1.wls)[4]
> tc = qt(0.975, df = df.residual(mod1.wls))
> uc.wls = b4.wls + tc * se_b4.wls
> lc.wls = b4.wls - tc * se_b4.wls
>
> # Robust GLS
> mod1.wls_hc = hccm(mod1.wls ,type = "hc1")
> vac.wls.HC1 = coeftest(mod1.wls, vcov.=mod1.wls_hc)
> b4.wls_rob = vac.wls.HC1[4,1]
> se_b4.wls_rob = vac.wls.HC1[4,2]
> uc_rob.wls = b4.wls_rob + tc * se_b4.wls_rob
> lc_rob.wls = b4.wls_rob - tc * se_b4.wls_rob
>
> cat(
+   "8.16(e)", "\n",
+   "the 95% CI using conventional GLS standard errors is : [",lc.wls,";",uc.wls,"]", "\n",
+   "the 95% CI using robust conventional GLS standard errors is : [",lc_rob.wls,";",uc_rob.wls,"]", "\n",
+   sep=""
+ )
8.16(e)
the 95% CI using conventional GLS standard errors is : [-119.8945,-33.71808]
the 95% CI using robust conventional GLS standard errors is : [-121.4134,-32.19919]

```

● Answer :

The 95% confidence interval using conventional GLS is [-119.8945, -33.71808], whereas the 95% confidence interval using robust GLS is [-121.4134, -32.19919]

與(a)、(d)小題的 95% confidence interval 相比，信賴區間變得比較窄，表示對於 kids 這個變數的估計精準度提高

